

Richard Bonichon

FORMAL METHODS & PROTOCOL ENGINEER · PHD

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- AUTOMATED THEOREM PROVING
- SMT SOLVING
- SYMBOLIC EXECUTION & BINARY ANALYSIS
- BFT CONSENSUS
- ZK PROOF SYSTEMS
- FUNCTIONAL PROGRAMMING (OCAML, RUST)
- LLM-ASSISTED DEVELOPMENT

Formal-methods researcher turned protocol engineer: automated theorem proving, SMT and binary-level verification (11 years at CEA, lead of BINSEC; 22 peer-reviewed papers), then BFT consensus and ZK protocol code in production.

WORK EXPERIENCE

2025-03	now	Senior Software Engineer · Repyh Labs Protocol Engineer - Documented the SDK and base libraries for early app developers - Modularized domain tasks; fixed transaction-application logic; typed abstractions - Added NFT support; designed and built the archive node - Built the integration-testing API and its test suite	Rust Rust Rust Rust
2022-09	2024-08	Software Engineer · O(1) Labs Crypto & Protocol Engineer - Codebase-wide cleanup, refactoring and higher test coverage - o1vm/Optimism: low-level comms with the fault-proof op-program - Berkeley hard-fork: pre/post-fork test harness - Prototyped account deletion (core zkApps feature)	OCaml Nix Rust Go Bash OCaml Dhall OCaml Postgresql
2021-01	2022-08	Software Engineer · Nomadic Labs Consensus Team Lead & Protocol Engineer - Test maintenance and migration Python → Text for the new consensus - Implemented Tenderbake consensus for the Ithaca protocol upgrade - Led the consensus team (from July 2021); Merge Team (core reviewers) member	Python OCaml OCaml
2019-09	2021-01	Senior Software Engineer · Tweag Project lead & Consultant - Tezos clients: Sapling, mockup, proxy - Gerrit plugins: owner approval, CI reporting - Built the internship program (setup, outreach, selection)	OCaml Rust Java JavaScript Go
2008-10	2019-08	Research Engineer · CEA Formal program analyses on source and binary code for safer, more secure software. Peer-reviewed papers at CAV, ICSE, CCS, ASE, RAID, ACSAC. - Lead dev & team co-lead of the open-source binary-analysis and symbolic-execution platform BINSEC - Co-advisor (day-to-day supervisor) of 2 PhD students: F. Recoules (inline-assembly verification, ICSE'21 Distinguished Paper) and M. D. Nguyen (binary-level directed fuzzing, RAID'20) - Metrics & Mthread Frama-C plug-ins - PathCrawler online : web UI for the concolic-testing tool - Lectures on software attacks & vulnerabilities (ENSTA)	OCaml Docker Python OCaml Shell Java JavaScript
2013-07	2015-06	Visiting Professor · DIMAp/UFRN Teaching (programming, formal methods) / Research (SMT solvers) - SMTpp : generic pre-processor & analyzer for SMT solvers - Portugol online interpreter - SMT solver veriT development	OCaml OCaml Javascript C OCaml
2007	2008	Post-doc · LORIA / Paréo Generation of construction functions respecting algebraic invariants for OCaml	
2006	2007	ATER (Teaching assistant) · Université Paris 6 Pierre & Marie Curie	

SELECTED PUBLICATIONS

Interface Compliance of Inline Assembly: Automatically Check, Patch and Refine

F. Recoules, S. Bardin, **R. Bonichon**, M. Lemerre, L. Mounier, M.-L. Potet

ICSE 2021, Distinguished Paper Award

Search-based Approaches for Local Blackbox Deobfuscation: Understand, Improve and Mitigate

G. Menguy, S. Bardin, **R. Bonichon**, C. de Souza Lima

CCS 2021

Binary-level Directed Fuzzing for Use-After-Free Vulnerabilities

M. D. Nguyen, S. Bardin, **R. Bonichon**, R. Groz, M. Lemerre

RAID 2020

How to Kill Symbolic Deobfuscation for Free (or: Unleashing the Potential of Path-Oriented Protections)

M. Ollivier, S. Bardin, **R. Bonichon**, J.-Y. Marion

ACSAC 2019

Get rid of inline assembly through verification-oriented lifting

F. Recoules, S. Bardin, **R. Bonichon**, L. Mounier, M.-L. Potet

ASE 2019

Model Generation for Quantified Formulas: A Taint-Based Approach

B. Farinier, S. Bardin, **R. Bonichon**, M.-L. Potet

CAV 2018

Zenon: An Extensible Automated Theorem Prover Producing Checkable Proofs

R. Bonichon, D. Delahaye, D. Doligez

LPAR 2007

TaMeD: A Tableau Method for Deduction Modulo

R. Bonichon

IJCAR 2004

Full list (23 publications): rbonichon.github.io/biblio.html 

EDUCATION

2003 | 2006 **PhD in Computer Science · Université Paris 6 Pierre & Marie Curie** 

Thesis: *Tableaux et déduction modulo* **Topics:** automated theorem proving, rewriting, cut elimination, logic

2002 | 2003 **MS in Computer Science (DEA) · Université Paris 7 Denis Diderot** 

Programming: Semantics, Proofs & Logic

1999 | 2002 **MS in Computer Science & Applied Mathematics · ENSEEIHT** 